Unlocking Business Insights from Real Estate Data

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Introduction

In today's data-driven world, businesses across diverse industries are leveraging the power of data analytics to gain a competitive edge. Real estate, a trillion-dollar industry, is no exception to this trend. This case study aims to demonstrate the significance of utilizing real estate data for data-driven decision-making. Using our dataset that contains information about property sales and the different features they have, we will be able to answer questions like what suburbs had the most sales, which real estate agents performed the best, and what features are most important in a house.

Finding and Using a Dataset

The Melbourne Housing Market[1] dataset we selected is a comprehensive collection of property transactions. This dataset contains attributes such as property addresses, suburbs, postcodes, property types, sale prices, transaction methods, agent details, and various property characteristics. The dataset was meticulously curated, ensuring it is a representation of the real estate market's intricacies.

Data Cleaning and Exploratory Data Analysis (EDA)

Before delving into the heart of our analysis, we recognized the paramount importance of data cleaning and exploratory data analysis (EDA). Data quality is fundamental to any data analysis, as poor data quality can compromise the integrity of the findings. The dataset was cleaned, addressing missing values, duplicates, and inconsistencies. We filled the "Bedroom2," "Bathroom," and "Car" columns with zeros, while rows with null values in critical columns like "Price" and "Postcode" were removed. Subsequently, we conducted EDA to unveil the dataset's underlying patterns, relationships, and trends. This process gave us a deep understanding of the data and allowed for more intricate analyses.

The "Date" column was converted to a datetime format to enable time-based analysis. Unnecessary columns were removed to streamline the dataset. We also ensured that numerical columns were turned into integers. We replaced values in the "Type" and "Method" columns to improve readability. Finally, we organized the columns for improved clarity. An image of our cleaned dataset is provided below.

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[1] Melbourne Housing Market dataset: https://kaggle.com/datasets/anthonypino/melbourne-housing-market/.

```
            Date            Address      Suburb  Postcode       Type  \
0     2016-12-03       85 Turner St  Abbotsford      3067      house
1     2016-02-04   25 Bloomburg St  Abbotsford      3067      house
2     2017-03-04       5 Charles St  Abbotsford      3067      house
3     2017-03-04  40 Federation La  Abbotsford      3067      house
4     2016-06-04       55a Park St  Abbotsford      3067      house
...          ...                ...         ...       ...        ...
27241 2018-02-24        13 Burns St  Yarraville      3013      house
27242 2018-02-24      29A Murray St  Yarraville      3013      house
27243 2018-02-24     147A Severn St  Yarraville      3013  townhouse
27244 2018-02-24   12/37 Stephen St  Yarraville      3013      house
27245 2018-02-24  3 Tarrengower St  Yarraville      3013      house

           Price      Method          Agent  Rooms  Bedroom2  Bathroom  Car
0      1480000.0        sold         Biggin      2         2         1    1
1      1035000.0        sold         Biggin      2         2         1    0
2      1465000.0  sold prior         Biggin      3         3         2    0
3       850000.0   passed in         Biggin      3         3         2    1
4      1600000.0  vendor bid         Nelson      4         3         1    2
...          ...         ...            ...    ...       ...       ...  ...
27241  1480000.0   passed in            Jas      4         4         1    3
27242   888000.0  sold prior        Sweeney      2         2         2    1
27243   705000.0        sold            Jas      2         2         1    2
27244  1140000.0  sold prior  hockingstuart      3         0         0    0
27245  1020000.0   passed in             RW      2         2         1    0

[27246 rows x 12 columns]
```

Cleaned Dataset

Identifying Business Value

While working with the dataset, we aimed to identify potential business value and actionable insights. Like many other industries, real estate faces various challenges, including understanding property pricing trends, evaluating the impact of different variables on property values, and assessing the effectiveness of various transaction methods. Real estate agents are also a factor that can be analyzed to see who performs best. We aimed to pinpoint these challenges and recognize where data could provide answers and solutions.

Data Analysis and Visualization

Our case study's central focus was on the data analysis and visualization phase. We harnessed the dataset's potential by applying a range of analytical and machine learning techniques. This involved assessing property pricing trends over time, identifying factors influencing property values, and scrutinizing the efficacy of different transaction methods. To communicate our findings effectively, we employed data visualization techniques, transforming complex data into easy-to-understand visualizations that answered our business questions. As shown from figure 1, you can clearly see the top 5 suburbs by property sales count. This provides insight into potential growing communities that will provide many business opportunities.

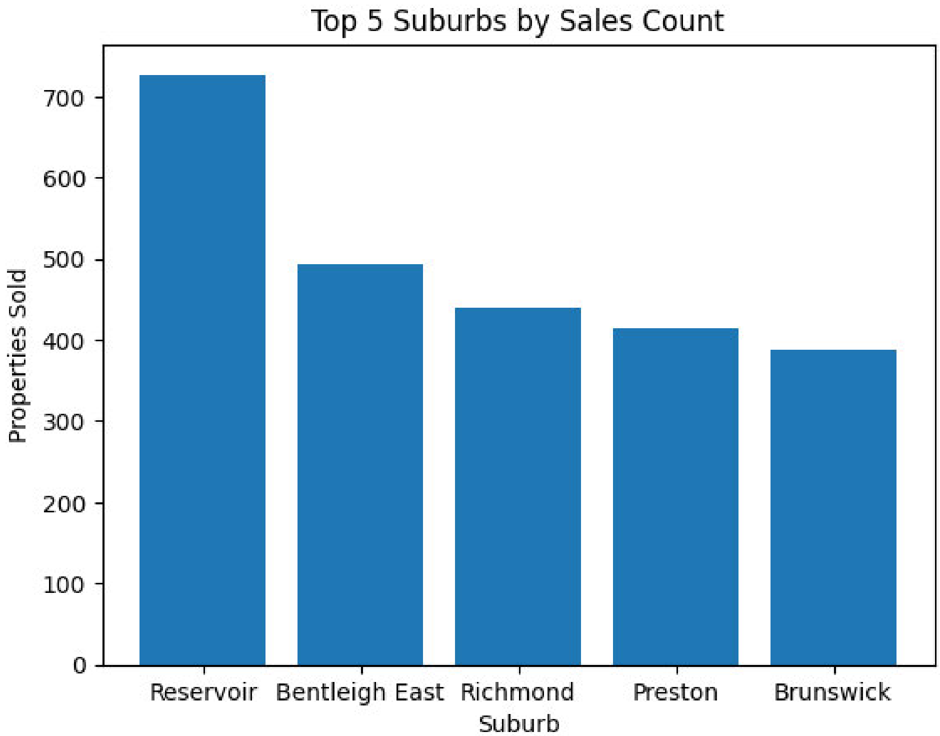

Figure 1

The two figures depict the top 10 real estate agents based on the number of sales, and the percentage of sales they have done compared to all sales. These figures allow us to see the most successful real estate agents, which is essential when determining who will try to sell more expensive properties when closing a sale is critical.

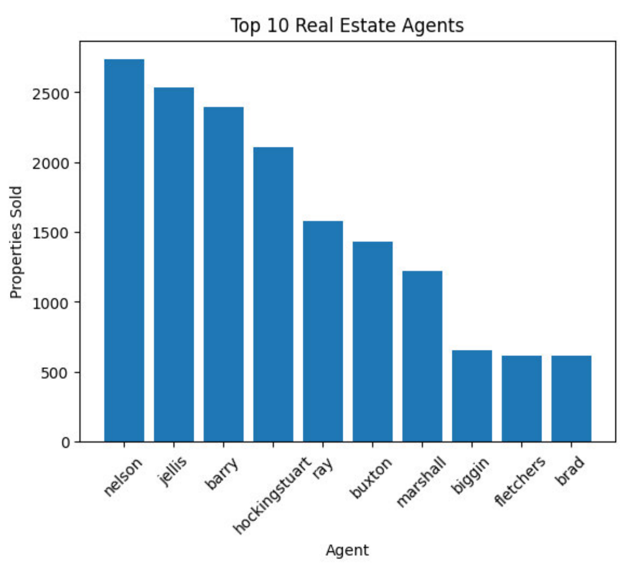

Figure 2

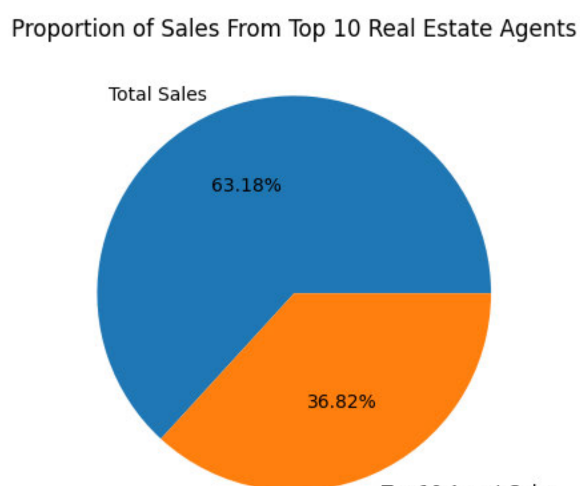

Figure 3

A time series analysis(figure 4) was conducted to demonstrate the change in average price over time. This provides critical insights into when properties sell for the highest values, indicating the best time for a real estate business to make money.

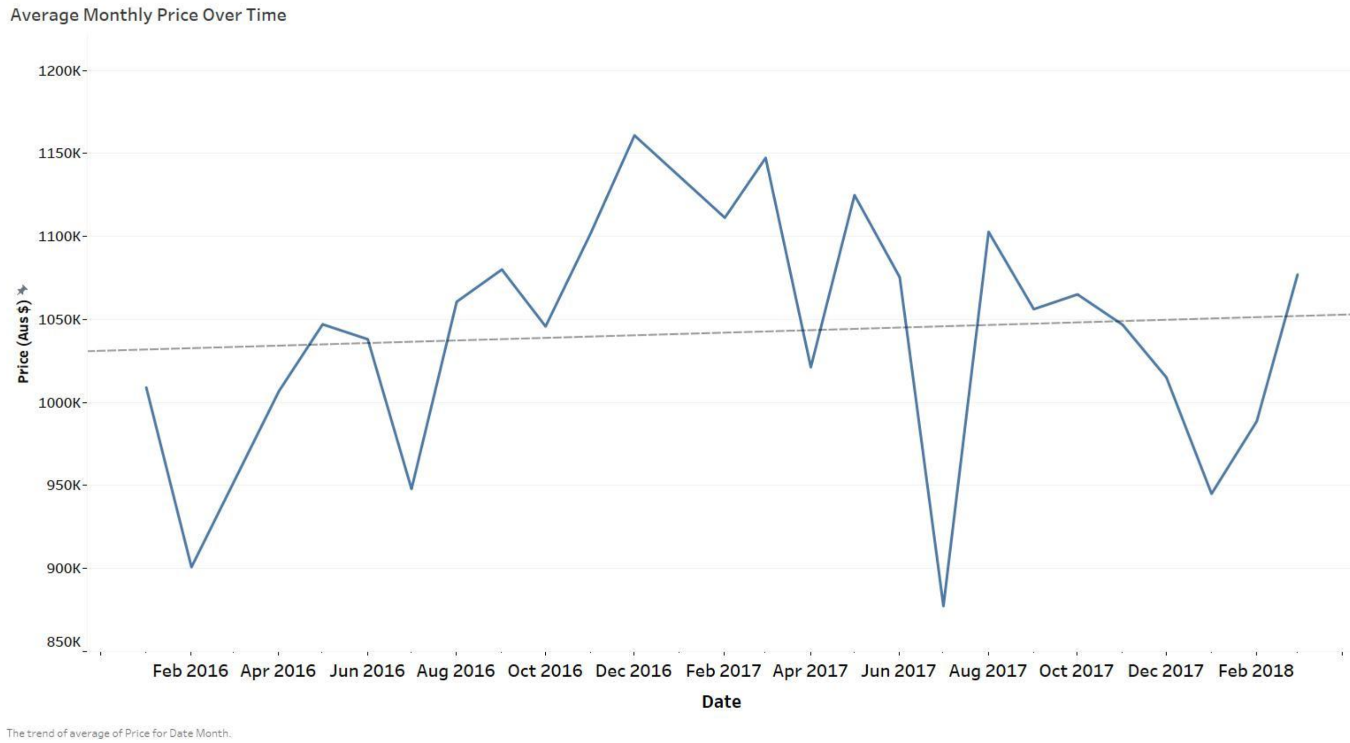

Figure 4

Finally, a correlation matrix was constructed to determine if features of a property are related to one another, and most importantly the price. The rooms variable is the number of total rooms in the property, while the Bedroom2 variable is the number of total bedrooms. The separation of the two allows us to determine if the number of total rooms is more correlated to the property value, or if only the number of bedrooms is most important. From the correlation matrix (figure 5) it is clear that price is mostly correlated to the number of rooms with a value of 0.47. A scatter plot (figure 6) was created to demonstrate these findings as well.

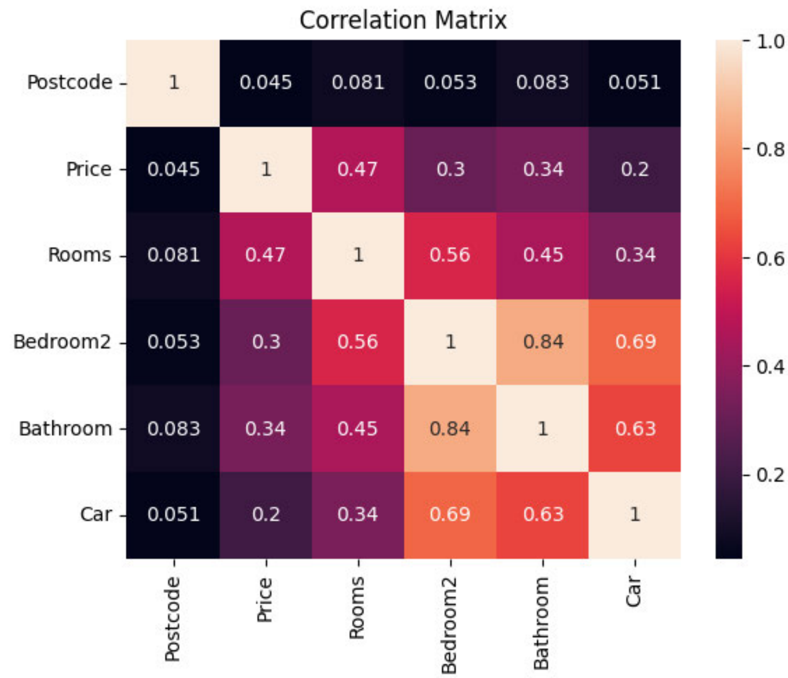

Figure 5

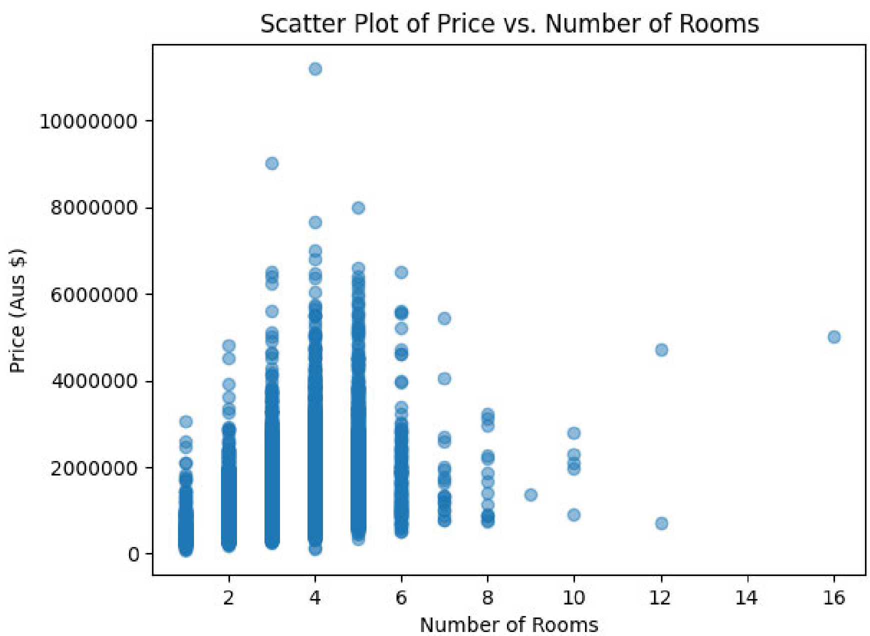

Figure 6

Output and Conclusion

With this refined data, the business can efficiently assess housing market trends, identify optimal investment opportunities, and make informed decisions regarding their agents. It enables stakeholders to answer critical questions such as:

1. **Market Trends:** By analyzing historical pricing data, businesses can gain insights into market trends. For example, it is possible to determine whether you should buy/sell a property in a specific neighborhood based on the number of bedrooms if these properties had the highest increase in price over a period of time.

2. **Optimal Investment**: The dataset empowers real estate investors to identify the suburbs with the ability to sell the most properties, reducing risks and maximizing profitability.

3. **Pricing Strategy:** For real estate agencies, the dataset offers guidance on setting competitive prices for properties based on historical sales data, ensuring fair and attractive listings.

4. **Agent Performance:** Businesses can optimize agent assignments and training programs by examining the effectiveness of various real estate agents based on their sales records.

5. **Property Attributes:** The dataset allows stakeholders to discern that the number of rooms correlates to a higher property value, and could potentially be used for machine learning or a regression analysis to predict property values.

6. **Market Entry Decisions:** For businesses contemplating entry into the Melbourne real estate market, the data serves as a valuable resource for making well-informed decisions regarding location, property type, and pricing.

Based on our analysis, our findings suggest that the best time to sell properties in the Melbourne area is in the months of December and January, with the worst time being in the months of June and July. Our trend line and regression model in figure 4 depicts that the market value is increasing every year. Focusing our top 10 sales agents on the top 5 suburbs in figures 1 and 2 will allow for us to sell more properties and make more money. The properties we should focus on are ones with a higher total count of rooms as this was determined to be the highest correlated feature when it comes to sales price. Our analysis of the highest average sale price during the year in combination of the top performing suburbs and real estate agents will lead to an increase in profit, especially if there is a focus on properties with a high amount of total rooms.

* This report was partially assisted by Chat-GPT.